CockpitSentinel: Distraction Model Ablation Report

Comparative Analysis: Baseline YOLO11s (v1) vs Merged Balanced Model (v2)

Ablation Finding: Fusing external driver phone data increased phone precision to 95.05% (+4.34%) and reduced box loss by -31.6%. However, domain shift lowered recall. The initial model (v1) remains the superior all-around production baseline with 88.55% F1-score.

1. Direct Metric Comparison (Benchmark Validation Set)

Metric	Version 1 (Initial Model)	Version 2 (Balanced Model)	Delta (Δ)	Analysis
Overall Precision	91.35%	88.79%	-2.56%	Clean detection fidelity
Overall Recall	85.91%	73.62%	-12.29%	v1 catches more infractions
Overall F1-Score	88.55%	80.49%	-8.06%	v1 has superior harmonic balance
Overall mAP@50	90.78%	84.33%	-6.45%	Higher overall accuracy in v1
Phone Precision	90.71%	95.05%	+4.34%	**v2 eliminated steering false-positives**
Phone Recall	86.67%	70.00%	-16.67%	v1 catches obscure phone angles
Smoking mAP@50	96.89%	89.71%	-7.18%	v1 remains exceptional on smoking
Validation Box Loss	1.9108	1.3059	-31.6%	**v2 has tighter bounding boxes**

2. Root-Cause Domain Discrepancy Analysis

1. **Camera Angle & Viewpoint:** The merged phone dataset (long-itwjh, 2,757 annotations) was captured from side-angle windshield positions, whereas our base dataset is positioned directly front-and-center.
2. **Annotation Bounding Style:** The external dataset strictly boxed the phone screen/chassis, while the base dataset boxed the driver's hand holding the phone. This competing supervision made the merged model more conservative.
3. **Convergence Depth:** Model v1 trained for 60 complete epochs with mosaic closed at epoch 50. Model v2 stopped early at epoch 36 via early-stopping.

3. Hardware Inference Latency (Intel Core i5 CPU)

Deployment Format	Engine	Latency (ms)	FPS	Speedup
PyTorch Checkpoint (best.pt)	PyTorch CPU	343.4 ms	2.9 FPS	1.00x
Model v1 OpenVINO (FP16)	OpenVINO CPU	44.0 ms	22.7 FPS	7.81x faster
Model v2 OpenVINO (FP16)	OpenVINO CPU	30.6 ms	32.6 FPS	**11.22x faster**